

SYDNEY BOYS' HIGH SCHOOL

MOORE PARK, SURRY HILLS



TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION 2000

MATHEMATICS

4 UNIT ADDITIONAL

*Time allowed — 3 Hours
(plus 5 minutes reading time)*

Examiner: C. Kourtesis

DIRECTIONS TO CANDIDATES

- *ALL* questions may be attempted.
- All necessary working should be shown in every question. Full marks may not be awarded for careless or badly arranged work.
- Approved calculators may be used.
- Start **each** section in a new booklet. Section A (questions 1, 2, 3), Section B (questions 4, 5, 6) and Section C (questions 7, 8).
- If required, additional booklets may be obtained from the Examination Supervisor upon request.

This is a trial paper and does not necessarily reflect the format or content of the HSC examination for this subject.

SECTION A

Question 1. (Start a new booklet)

15 Marks

(a) If $z = (1 - i)^{-1}$

5

(i) Express $\bar{z}$ in modulus-argument form,(ii) If $(\bar{z})^{13} = a + ib$ where a and b are real numbers, find the values of a and b .(b) Find the cartesian equation of the locus of a point P which represents the complex number z where

2

$$|z - i| = |z|$$

(c) Sketch the region in the complex plane where

3

$$\operatorname{Re}[(2 - 3i)z] < 12$$

(d) (i) On an Argand diagram sketch the locus of a point P , corresponding to the complex number z , where

3

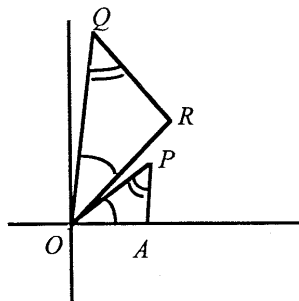
$$|z - 3| = 3$$

(ii) Use your diagram in (i) to explain why

$$\arg(z - 3) = \arg z^2$$

(e)

2



The points A , P and R in the complex plane correspond to the complex numbers 1 , $\frac{3}{2} + i$ and $2 + 2i$ respectively. Triangles OAP and ORQ are similar with corresponding angles as indicated.

Find the complex number represented by Q .

Question 2.

15 Marks

(a) Find $\int \frac{dx}{x^2 - 4x + 9}$ 2

(b) (i) Express $\frac{4x-2}{(x^2-1)(x-2)}$ in the form $\frac{Ax+B}{x^2-1} + \frac{C}{x-2}$, where A , B and C are constants. 5

(ii) Hence evaluate

$$\int_3^6 \frac{4x-2}{(x^2-1)(x-2)} dx$$

(c) Find $\int \frac{e^{2x}}{e^x-1} dx$ by using the substitution $u = e^x$. 3

(d) (i) If $u_n = \int_0^{\frac{\pi}{2}} \theta \sin^n \theta d\theta$ where $n \geq 1$, use integration by parts to prove that 5

$$u_n = \frac{n-1}{n} u_{n-2} + \frac{1}{n^2}$$

(ii) Hence show that $u_5 = \frac{149}{225}$

Question 3.

15 Marks

- (a) The polynomial
- $P(z)$
- has equation

3

$$P(z) = z^4 - 2z^3 - z^2 + 2z + 10$$

Given that $z - 2 + i$ is a factor of $P(z)$, express $P(z)$ as a product of two real quadratic factors.

- (b) The remainder when
- $x^4 + ax + b$
- is divided by
- $(x - 2)(x + 1)$
- is
- $x + 2$
- . Find the values of
- a
- and
- b
- . 2

- (c) (i) Show that

10

$$\tan 3\theta = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$$

- (ii) Find the general solution of the equation
- $\tan 3\theta = \sqrt{3}$

- (iii) Using the substitution
- $x = \tan \theta$
- , express the equation in (ii) as a polynomial equation in terms of
- x
- .

- (iv) Hence show that
- $\tan \frac{\pi}{9} + \tan \frac{4\pi}{9} + \tan \frac{7\pi}{9} = 3\sqrt{3}$

- (v) Find the polynomial of least degree that has zeros

$$\left(\cot \frac{\pi}{9} \right)^2, \left(\cot \frac{4\pi}{9} \right)^2, \left(\cot \frac{7\pi}{9} \right)^2$$

SECTION B

Question 4. (Start a new booklet)

15 Marks

(a) Consider the function

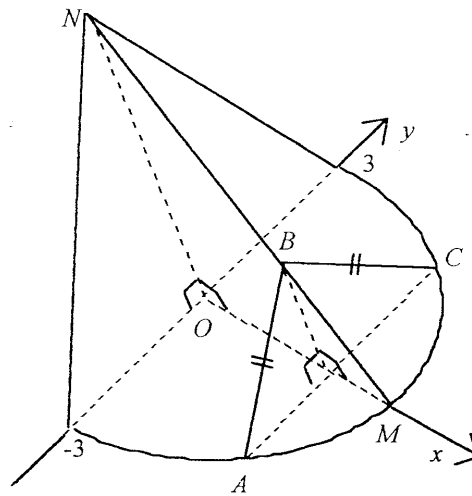
9

$$F(x) = \left(\frac{x+4}{x} \right)^2, \quad x \neq 0$$

- (i) Find all the turning points of $y = F(x)$,
- (ii) Determine the coordinates of the point of inflexion,
- (iii) Find the equations of any asymptotes,
- (iv) Sketch the curve $y = F(x)$ for all points in its domain.

(b)

6



A solid figure has a semi circular base of radius 3 cm. Cross sections taken perpendicular to the x axis are isosceles triangles. The vertical cross section containing the radius OM of the base of the solid is a right isosceles triangle OMN , where $OM = ON$.

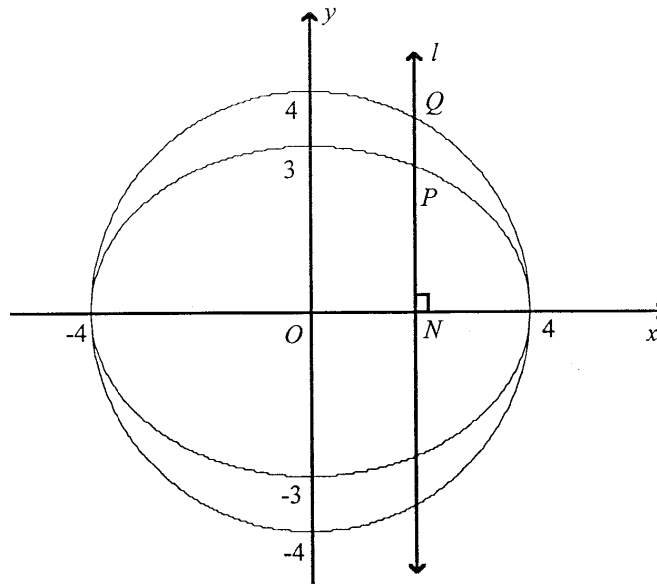
- (i) Show that the area, A , of triangle ABC (where $AB = BC$) is given by

$$A = (3-x)(9-x^2)^{\frac{1}{2}}$$

- (ii) Find the volume of the solid.

(a)

13



The diagram shows the ellipse, E , with equation $\frac{x^2}{16} + \frac{y^2}{9} = 1$ and its auxiliary circle C . The coordinates of a point P on E are $(4\cos\theta, 3\sin\theta)$.

A straight line, l , parallel to the y axis intersects the x axis at N and the curves E and C at the points P and Q respectively.

- (i) Find the eccentricity of E ,
 - (ii) Write down the coordinates of N and Q ,
 - (iii) Find the equations of the tangents at P and Q to the curves E and C respectively,
 - (iv) The tangents at P and Q intersect at a point R . Show that R lies on the x axis,
 - (v) Prove that $ON \cdot OR$ is independent of the positions of P and Q .
- (b) State whether the following is True or False. Give brief reasons.

2

Note: You are NOT required to find the primitive function.

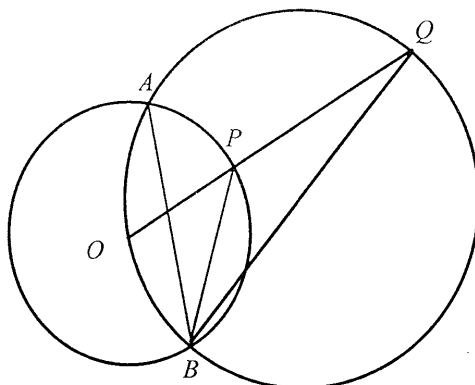
$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \tan^9 \theta \, d\theta > 0$$

Question 6.

15 Marks

(a)

5



In the diagram above, the centre O of the small circle APB lies on the circumference of the larger circle AQB . The points O, P and Q are collinear.

Prove that BP bisects $\angle ABQ$

- (b) (i) Sketch the region in the number plane that contains all points satisfying simultaneously the inequalities

6

$$x \leq 1, y \geq 1 \text{ and } y \leq e^x$$

- (ii) This region is rotated through one complete revolution about the x axis. Use the method of cylindrical shells to show that the volume of the resulting solid is

$$\frac{\pi}{2}(e^3 - 3)$$

- (c) If a function $f(x)$ is continuous for $a \leq x \leq b$

4

- (i) Show that $\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$

- (ii) Hence prove that

$$\left| \int_0^\pi 4^x \cos x dx \right| \leq \frac{2^{2\pi} - 1}{2 \ln 2}$$

SECTION C

Question 7. (Start a new booklet)

15 Marks

- (a) A particle of mass m is projected vertically upwards in a medium where it experiences a resistance of magnitude mkv^2 where k is a positive constant and v is the velocity of the particle. 11

During the downward motion the terminal velocity of the particle is V . It's initial velocity of projection is $\frac{1}{3}$ of this terminal velocity.

- (i) By considering the forces on the particle during its downward motion, show that

$$kV^2 = g$$

(where g is the acceleration due to gravity)

- (ii) Show that during its upward motion the acceleration of the particle $\ddot{x}$ is given by

$$\ddot{x} = -g \left(1 + \frac{v^2}{V^2} \right)$$

- (iii) If the distance travelled by the particle in its upward motion is x when its velocity is v , show that the maximum height H reached is given by

$$H = \frac{V^2}{2g} \ln \left(\frac{10}{9} \right)$$

- (iv) The velocity of the particle is v when it has fallen a distance y from its maximum height. Show that

$$y = \frac{V^2}{2g} \ln \left[\frac{V^2}{V^2 - v^2} \right]$$

- (v) The velocity of the particle is U when it returns to its point of projection. Show that

$$\frac{V}{U} = 10^{\frac{1}{2}}$$

- (b) (i) From 11 distinct consonants and 5 distinct vowels, how many words can be formed, each containing 5 distinct consonants and 3 distinct vowels? 4
- (ii) In how many ways is it possible to allocate 6 people to 3 different courts in a singles tennis tournament?

Question 8.

15 Marks

- (a) (i) Show that $\frac{a+b}{2} \geq \sqrt{ab}$ for all positive numbers a and b .

4

- (ii) If a, b, c and d are positive numbers prove that

$$4(ab + bc + cd + ad) \leq (a + b + c + d)^2$$

- (b) If u and v are real numbers such that $u + v \neq 0$ and $v \neq 0$,

5

- (i) Show that if there is only one real root of the equation $x^2 + ux + v = 0$ (where $0 < x < 1$) then

$$v(1 + u + v) < 0$$

- (ii) Hence, or otherwise, prove that the equation

$$\frac{1}{x+2} + \frac{u}{x+1} + \frac{v}{x} = 0$$

has only one positive root.

- (c) Given the function $f(x) = x^n e^{-x}$ where n is a positive integer and $x > 0$:

6

- (i) Prove that there is only one turning point and that this occurs at $x = n$. Deduce that it is a maximum turning point.

- (ii) Sketch the graph of $y = f(x)$,

- (iii) By considering the values of $f(n)$, $f(n-1)$ and $f(n+1)$ prove that

$$\left(1 + \frac{1}{n}\right)^n < e < \left(1 - \frac{1}{n}\right)^{-n}$$

THIS IS THE END OF THE PAPER.



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**TRIAL HIGHER SCHOOL
CERTIFICATE EXAMINATION**

Mathematics Extension 2

Sample Solutions

Question 1

(a) $z = \frac{1}{1-i}$

$= \frac{1}{2}(1+i)$

(i) $\bar{z} = \frac{1}{2}(1-i)$

$= \frac{1}{2} \text{cis}(-\frac{\pi}{4})$

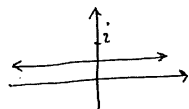
(ii) $(\bar{z})^{13} = (\frac{1}{2})^{13} \text{cis}(-\frac{13\pi}{4})$

$= \frac{1}{64\sqrt{2}} \text{cis} \frac{3\pi}{4}$

$= \frac{1}{64\sqrt{2}} (-\frac{1}{2} + \frac{i}{2})$

$\therefore a = -\frac{1}{64\sqrt{2}}, b = \frac{1}{64\sqrt{2}}$

(b) $|z-i| = |z|$



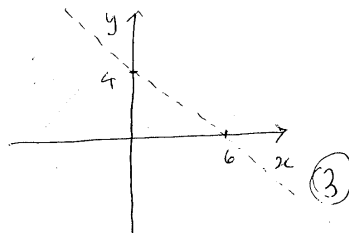
$y = \frac{1}{2}$

(c) $\text{Re}[(z-3i)z] < 12$

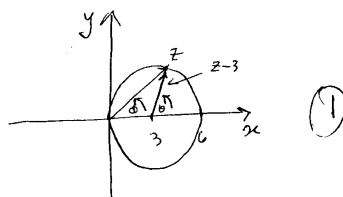
$\therefore \text{Re}[(2-3i)(x+iy)] < 12$

$\text{Re}[(2x+3y)+i(-3x+2y)] < 12$

$2x+3y < 12$



(d) (i) $|z-3| = 3$



(ii) RTP $\arg(z-3) = \arg z^2$

In the diagram let $\theta = \arg(z-3)$ and $\phi = \arg z$

Now $\theta = 2\phi$ (angle at centre is double angle at circumference)

$\therefore \arg(z-3) = 2\arg z$
 $= \arg z^2$

(e) Let $\angle OPQ = \theta \therefore \angle OQP = \theta$ (given)
Let $\angle POQ = \phi$

Now $\arg \vec{OQ} = \phi + \theta$
 $= \arg \vec{OP} + \arg \vec{OQ}$
 $= \arg(\vec{OP} \times \vec{OQ})$

Also $\frac{\vec{OQ}}{\vec{OP}} = \frac{\vec{OP}}{1}$

$\therefore \vec{OQ} = \vec{OP} \times \vec{OP}$

$\therefore |\vec{OQ}| = |\vec{OP}| \cdot |\vec{OP}|$

$\therefore \vec{OQ} = \vec{OP} \cdot \vec{OP}$
 $= (2+2i)(\frac{3}{2}+i)$
 $= 3+2i+3i-2$
 $= 1+5i$

Question 2

(a) $\int \frac{dx}{x^2+4x+9}$
 $= \int \frac{dx}{(x+2)^2+5}$

$= \frac{1}{\sqrt{5}} \tan^{-1}(\frac{x+2}{\sqrt{5}}) + C$

(b) $\frac{4x-2}{(x^2-1)(x-2)} = \frac{Ax+B}{x^2-1} + \frac{C}{x-2}$

$\therefore 4x-2 = (x-2)(Ax+B) + (x^2-1)C$
 $= Ax^2+Bx-2Ax-2B+x^2C-C$
 $= (A+C)x^2+(B-2A)x-2B-C$

Equating Coefficients:

$x^2: 0 = A + C$

$x: 4 = -2A + B$

$x^0: -2 = -2B - C$

$\text{①} + \text{③}: -2 = A - 2B$

$\text{②} + 2 \times \text{③}: 0 = -3B$

$\therefore B = 0; A = -2, C = 2$

$\therefore \frac{4x-2}{(x^2-1)(x-2)} = \frac{-2x}{x^2-1} + \frac{2}{x-2}$

$\therefore \int_3^6 \frac{4x-2}{(x^2-1)(x-2)} dx = \int_3^6 (\frac{2}{x-2} - \frac{2x}{x^2-1}) dx$

$= [2\ln(x-2) - \ln(x^2-1)]_3^6$
 $= [(2\ln 4 - \ln 35) - (2\ln 1 - \ln 8)]$
 $= 2\ln 4 - \ln 35 + \ln 8$
 $= 7\ln 2 - \ln 35$

(c) $I = \int \frac{e^{2x}}{e^{2x}-1} dx$ let $u = e^x$
 $du = e^x dx$

$= \int \frac{u du}{u^2-1}$

$= \int \frac{u-1+1}{u^2-1} du$

$= \int (1 + \frac{1}{u-1}) du$

$= u + \ln(u-1) + C$

$= e^x + \ln(e^x-1) + C$

(d) $u_n = \int_0^{\pi/2} \theta \sin^n \theta d\theta$

$= \int_0^{\pi/2} \theta \sin^{n-1} \theta \cdot \sin \theta d\theta$

Let $u = \theta \sin^{n-1} \theta$ $v' = \sin \theta$

$= [-\cos \theta \cdot \theta \sin^{n-1} \theta]_0^{\pi/2} + \int_0^{\pi/2} \cos \theta (\theta \sin^{n-1} \theta)' d\theta$
 $+ \sin^{n-1} \theta d\theta$

$= 0 + \int_0^{\pi/2} \cos \theta \sin^{n-1} \theta d\theta + \int_0^{\pi/2} (\theta \sin^{n-2} \theta \cos \theta + \sin^{n-2} \theta \sin \theta) d\theta$

$= \frac{1}{n} [\sin^n \theta]_0^{\pi/2} + (\theta) \int_0^{\pi/2} \sin^{n-2} \theta d\theta + \sin^{n-2} \theta d\theta$

$= \frac{1}{n} + (n-1) u_{n-2} - (n-1) u_n$

$\therefore u_n + (n-1) u_n = (n-1) u_{n-2} + \frac{1}{n}$
 $\therefore u_n = \frac{n-1}{n} u_{n-2} + \frac{1}{n}$

Q2 (Contd)

(a) Contd

$$(ii) u_5 = \frac{1}{25} + \frac{4}{5} u_3$$

$$= \frac{1}{25} + \frac{4}{5} \left(\frac{1}{9} + \frac{2}{3} u_1 \right) (*)$$

$$u_1 = \int_0^{\pi/2} \theta \sin \theta d\theta$$

$$= \int_0^{\pi/2} \theta \cdot d(-\cos \theta) d\theta$$

$$= [\theta(-\cos \theta)]_0^{\pi/2} + \int_0^{\pi/2} \cos \theta d\theta$$

$$= 0 + [\sin \theta]_0^{\pi/2}$$

$$= 1$$

Substituting into (*)

$$u_5 = \frac{1}{25} + \frac{4}{5} \left(\frac{1}{9} + \frac{2}{3} \right)$$

$$= \frac{149}{225}$$

Question 3

(a) $P(z) = z^4 - 2z^3 - z^2 + 2z + 10$

$z - 2 + i$ is a factor

ie $z = 2 - i$ is root

$\therefore z = 2 + i$ is also a root
(Conjugate root theorem)

$\therefore (z - 2 + i)(z - 2 - i)$ is a factor

ie $z^2 - 4z + 5$ is a factor

By long division

$$P(z) = (z^2 - 4z + 5)(z^2 + 2z + 2)$$

(b) By long division, the remainder

$$x+2 \equiv (a+5)x + (b+6)$$

$$\therefore 1 = a+5 \quad 2 = b+6$$

$$a = -4 \quad b = -4$$

(c) (i) $\tan 3\theta = \tan(2\theta + \theta)$

$$= \frac{\tan 2\theta + \tan \theta}{1 - \tan 2\theta \tan \theta}$$

$$= \frac{2 \tan \theta}{1 - \tan^2 \theta} + \tan \theta$$

$$= \frac{2 \tan \theta + \tan \theta - \tan^3 \theta}{1 - \tan^2 \theta}$$

$$= \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$$

(ii) $\tan 3\theta = \sqrt{3}$

$$3\theta = \frac{\pi}{3} + k\pi \quad k=0, \pm 1, \pm 2$$

$$\therefore \theta = \frac{\pi}{9} + \frac{k\pi}{3}$$

$$= \frac{\pi}{9} (3k+1)$$

$$\theta = \frac{\pi}{9}, \frac{4\pi}{9}, \frac{7\pi}{9}, \text{ etc }$$

(iii) $\tan 3\theta = \sqrt{3}$

$$\therefore \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta} = \sqrt{3}$$

Put $x = \tan \theta$

Q3 (i) (iii) Contd

$$\therefore 3x - x^3 = \sqrt{3} - 3\sqrt{3}x^2$$

$$\therefore x^3 - 3\sqrt{3}x^2 - 3x + \sqrt{3} = 0 (*)$$

(iv) $\tan \frac{\pi}{9}, \tan \frac{4\pi}{9}, \tan \frac{7\pi}{9}$
are all ^{distinct} roots of (*) since
 $x = \tan \theta$

Hence sum of roots

$$-\frac{b}{a} = 3\sqrt{3}$$

$$= \tan \frac{\pi}{9} + \tan \frac{4\pi}{9} + \tan \frac{7\pi}{9}$$

(v) let roots of
 $x^3 - 3\sqrt{3}x^2 - 3x + \sqrt{3} = 0$ are
 α, β, γ then the reciprocal
roots are $\frac{1}{\alpha} = \cot^2 \frac{\pi}{9}$

$$\frac{1}{\beta} = \cot^2 \frac{4\pi}{9}$$

$$\frac{1}{\gamma} = \cot^2 \frac{7\pi}{9}$$

Now let $y = \frac{1}{x^2}$

$\therefore d = \sqrt{\frac{1}{y}}$, but since

d is root of the above eqn
it follows that

$$\sqrt{\frac{1}{y}}^3 - 3\sqrt{3}(\sqrt{\frac{1}{y}})^2 - 3\sqrt{\frac{1}{y}} + \sqrt{3} = 0$$

$$\therefore \sqrt{\frac{1}{y}}^3 - 3\sqrt{\frac{1}{y}} = 3\sqrt{3} \cdot \frac{1}{y} - \sqrt{3}$$

Multiply both sides

$$\left(\frac{1}{y}\right)^3 - 6\left(\frac{1}{y}\right)^2 + \frac{27}{4y} - \frac{18}{y} + 3$$

(*)

Multiply both sides by y^3 :

$$1 - 6y + 9y^2 = 27y - 18y^2 + 3y^3$$

$\therefore$ Required eqn is:

$$3y^3 - 27y^2 + 33y - 1 = 0$$

Put x for y :

$$3x^3 - 27x^2 + 33x - 1 = 0$$

Section B.

$$4.) \quad a) \quad y = \frac{(x+4)^2}{x^2} - \left(1 + \frac{4}{x}\right)^2$$

$$= \frac{x^2 + 8x + 16}{x^2}$$

$$= 1 + \frac{8}{x} + \frac{16}{x^2}$$

$$ii) \quad y' = -\frac{8}{x^2} - \frac{32}{x^3}$$

$$= -\frac{(8x+32)}{x^3} = -\frac{8(x+4)}{x^3}$$

$$y' = 0 \text{ when } x = -4 \quad (-4, 0) \checkmark$$

∴ minimum ✓

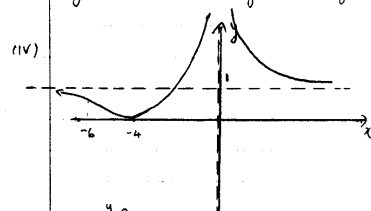
x	-4
y'	0

$$ii) \quad y'' = \frac{16}{x^3} + \frac{96}{x^4} = \frac{16x+96}{x^4} \checkmark$$

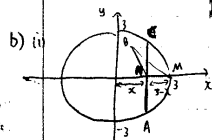
$$y'' = 0 \Rightarrow x = -6. \quad \therefore y = \frac{4}{36} = \frac{1}{9}$$

$$\therefore \text{POI } (-6, \frac{1}{9}) \checkmark$$

iii) $x = 0$ is the vertical asymptote. ✓
 $y = 1$ is the horizontal asymptote. ✓



For $x > 0$, $y' < 0$



$$AC = 2y = 2(9-x^2)^{\frac{1}{2}} \quad \boxed{BN = 8/4 = 3-x}$$

$$\Delta ABC = \frac{1}{2} \times (3-x) \times 2(9-x^2)^{\frac{1}{2}}$$

$$= (3-x)(9-x^2)^{\frac{1}{2}} \checkmark$$

$$V = \int_0^3 (3-x)(9-x^2)^{\frac{1}{2}} dx \checkmark$$

$$= 3 \int_0^3 \sqrt{9-x^2} dx + \int_0^3 -x \sqrt{9-x^2} dx \checkmark$$

$$V = 3 \times \frac{9\pi}{4} + \frac{1}{2} \int_0^3 -2x \sqrt{9-x^2} dx$$

$$= \frac{27\pi}{4} + \frac{1}{2} \int_9^0 \sqrt{u} du$$

$$= \frac{27\pi}{4} + \frac{1}{2} \left[\frac{2}{3} u^{3/2} \right]_9^0$$

$$= \frac{27\pi}{4} + \frac{1}{3} (0 - 27)$$

$$= \frac{27\pi}{4} - 9 \checkmark \checkmark$$

$$u = 9-x^2$$

$$du = -2x dx$$

$$x=0, u=9$$

$$x=3, u=0$$

$$\int u^{\frac{1}{2}} du = \frac{2}{3} u^{3/2}$$

5)

a) $\frac{x^2}{16} + \frac{y^2}{9} = 1$

$a=4, b=3$

(ii) $e^2 = 1 - \frac{b^2}{a^2} = 1 - \frac{9}{16} = \frac{7}{16}$

$\therefore e = \frac{\sqrt{7}}{4}$ ✓

(iii) N (4cosθ, 0) ✓
Q (4cosθ, 4sinθ) ✓

(iii) P: $\frac{2x}{16} + \frac{2y}{9} \frac{dy}{dx} = 0$

$\therefore \frac{dy}{dx} = -\frac{9x}{16y} = -\frac{9(4\cos\theta)}{16(3\sin\theta)} = -\frac{3\cos\theta}{4\sin\theta}$ ✓

$\therefore y - 3\sin\theta = \frac{-3\cos\theta}{4\sin\theta} (x - 4\cos\theta)$ ✓

$\therefore 4y\sin\theta - 12\sin^2\theta = -3x\cos\theta + 12\cos^2\theta$

$\therefore 3x\cos\theta + 4y\sin\theta = 12$ ✓
 $\boxed{\frac{x\cos\theta}{4} + \frac{y\sin\theta}{3} = 1}$ ✓

Q: $x^2 + y^2 = 16$

$2x + 2y \frac{dy}{dx} = 0$

$\therefore \frac{dy}{dx} = -\frac{x}{y} = -\frac{4\cos\theta}{4\sin\theta} = -\frac{\cos\theta}{\sin\theta}$ ✓

$y - 4\sin\theta = \frac{-\cos\theta}{\sin\theta} (x - 4\cos\theta)$ ✓

$y\sin\theta - 4\sin^2\theta = -x\cos\theta + 4\cos^2\theta$

$\boxed{x\cos\theta + y\sin\theta = 4}$ ✓

(iv) $3x\cos\theta + 4y\sin\theta = 12$ — (1)

$x\cos\theta + y\sin\theta = 4$ — (2)

(2) × 3

$3x\cos\theta + 4y\sin\theta = 12$
 $3x\cos\theta + 3y\sin\theta = 12$] -

$\therefore y\sin\theta = 0$
 $y = 0$ (sinθ ≠ 0) ✓

$\therefore 3x\cos\theta = 12$

$\therefore x\cos\theta = 4$

$\therefore R(4\sec\theta, 0)$ ✓ which lies on the x-axis.

N.B. If θ = π then P and Q co-incide on the x-axis anyway

(v) ON = |4cosθ|

OR = |4secθ| ✓

$\therefore ON \cdot OR = 16$ ✓ which is independent of θ, thus independent of P and Q.

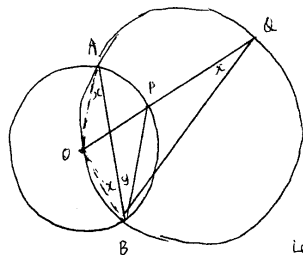
(b) $\int_{-\pi/4}^{\pi/4} \tan^q \theta d\theta > 0$

False: $\tan\theta$ is odd, so $\tan^q\theta$ is odd and continuous for $-\frac{\pi}{4} \leq \theta \leq \frac{\pi}{4}$ ✓

$\boxed{\int_{-a}^a f(x) dx = 0 \text{ for odd } f(x)}$

$\int_{-\pi/4}^{\pi/4} \tan^q \theta d\theta = 0$

6 a)



let $x = \angle OAB$

In circle AQB
 $\angle OAB = x$ ✓ (same segment as $\angle OAB$)

$OA = OB$

$\therefore \triangle OAB$ is isosceles

$\therefore \angle OBA = x$ ✓

let $y = \angle ABP$

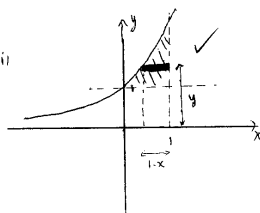
$\therefore \angle OPB = x+y$ ✓ ($\triangle OPB$ is isosceles)

$\therefore \angle BPQ = 180 - (x+y)$ ✓

$\therefore \angle PBQ = y$ (angle sum of $\triangle PBQ$) ✓

$\therefore PB$ bisects $\angle ABQ$

b) (ii)



$\int y \ln y dy$ $u = y$ $u' = \frac{1}{y}$
 $v = \ln y$ $v' = \frac{1}{y}$

$= \frac{1}{2} y^2 \ln y \Big|_1^e - \int_1^e \frac{1}{2} y dy$

$= \frac{e^2}{2} - \left[\frac{y^2}{4} \right]_1^e$

$= \frac{e^2}{2} - \frac{e^2}{4} + \frac{1}{4}$

$= \frac{e^2 + 1}{4}$ ✓

(iii) $\Delta A \div 2\pi y(1-x)$ ✓

$\Delta v \div 2\pi y(1-x) \Delta y$

$v = 2\pi \int_1^e y(1-x) dy$ ✓

$= 2\pi \int_1^e y(1-\ln y) dy$ ✓

$= 2\pi \left[\int_1^e y dy - \int_1^e y \ln y dy \right]$

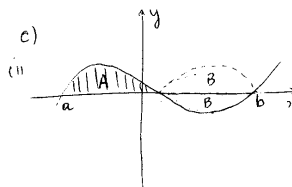
$= 2\pi \left[\frac{y^2}{2} \Big|_1^e - \left(\frac{e^2}{4} + \frac{1}{4} \right) \right]$

$= 2\pi \left[\frac{e^2}{2} - \frac{1}{2} - \frac{e^2}{4} - \frac{1}{4} \right]$

$= 2\pi \left[\frac{e^2}{4} - \frac{3}{4} \right] = 2\pi \left[\frac{e^2-3}{4} \right]$

$= \frac{\pi}{2} [e^2-3]$ ✓

QED



III $B, A = \text{areas} (>0)$

LHS = $|A-B| = \left| \int_a^b f(x) dx \right|$

RHS = $A+B = \int_a^b |f(x)| dx$

$|f(x)|$ is always positive so $\int_a^b |f(x)| dx$ is a positive value.

Given that $f(x)$ can cross the x-axis for $a \leq x \leq b$

then $\int_a^b f(x) dx$ may involve subtraction ✓

Equality if $f(x) \geq 0$ for $a \leq x \leq b$

$\therefore \left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$

(ii) $|\cos x| \leq 1$ $\left| \int_0^\pi 4^x \cos x dx \right| \leq \int_0^\pi |4^x \cos x| dx$ from (i)

$\leq \int_0^\pi 4^x dx$ ✓

$= \left[\frac{4^x}{\ln 4} \right]_0^\pi$ ✓

$= \frac{2^x}{2 \ln 2} \Big|_0^\pi$

$= \frac{2^{\pi}-1}{2 \ln 2}$ QED ✓

$\int a^x dx = \frac{a^x}{\ln a} + C$

7(a) $\uparrow mkv^2$
 $\downarrow mg$
 (i) $\ddot{x} = g - kv^2$
 At terminal velocity, $\ddot{x} = 0$
 $g - kv^2 = 0$
 $kv^2 = g$ 2

(ii) $\ddot{x} = -(g + kv^2)$
 $= -(g + \frac{g}{v^2} x^2)$
 $= -g(1 + \frac{x^2}{v^2})$

(iii) $v \frac{dv}{dx} = -g(1 + \frac{x^2}{v^2})$
 $v \frac{dv}{dx} = -\frac{g}{v^2}(v^2 + x^2)$
 $\int v \frac{dv}{v^2 + x^2} = -\int \frac{g}{v^2} dx$

$\frac{1}{2} \ln(v^2 + x^2) = -\frac{g}{v^2} x + c$
 When $x=0$: $\frac{1}{2} \ln(v^2 + \frac{v^2}{g}) = c$
 $\frac{g}{v^2} x = \frac{1}{2} \ln(\frac{10v^2}{g}) - \frac{1}{2} \ln(v^2 + x^2)$

At max ht, $v=0$
 $\frac{g}{v^2} H = \frac{1}{2} \ln(\frac{10v^2}{g})$
 $H = \frac{v^2}{2g} \ln(\frac{10}{g})$ 3

(iv) $\ddot{y} = g - kv^2$
 $v \frac{dv}{dy} = g - kv^2$
 $\int \frac{v dv}{g - kv^2} = \int dy$
 $-\frac{1}{2k} \ln(g - kv^2) = y + c$
 When $y=0$ $v=0$

$-\frac{1}{2k} \ln g = c$
 $y = \frac{1}{2k} \ln \frac{g}{g - kv^2}$
 $= \frac{1}{2k} \ln \frac{g}{g - \frac{g}{v^2} v^2}$
 $= \frac{v^2}{2g} \ln \frac{v^2}{v^2 - v^2}$ 3

(v) When $y = \frac{v^2}{2g} \ln(\frac{10}{g})$:
 $\frac{v^2}{2g} \ln(\frac{10}{g}) = \frac{v^2}{2g} \ln \frac{v^2}{v^2 - v^2}$
 $\frac{10}{g} = \frac{v^2}{v^2 - v^2}$
 $10v^2 - 10v^2 = 9v^2$
 $v^2 = 10v^2$
 $\frac{v^2}{v^2} = 10$ 2
 $\frac{v}{v} = 10^{\frac{1}{2}}$

(b) (i) No. of words
 $= {}^{11}C_5 \times {}^5C_3 \times 8!$
 chooses consonants chooses vowels mix them up
 $(= 186\ 278\ 400)$ 2

(ii) No. of ways
 $= {}^6C_2 \times {}^4C_2$
 choose 2 for the first court choose 2 for the 2nd court
 $= 90$ 2

8 (a) (i) $(\sqrt{a} - \sqrt{b})^2 \geq 0$
 $a - 2\sqrt{ab} + b \geq 0$
 $a + b \geq 2\sqrt{ab}$
 $\frac{a+b}{2} \geq \sqrt{ab}$ 2

(ii) From (i)
 $\frac{(a+c)(b+d)}{2} \geq \sqrt{(a+c)(b+d)}$
 $((a+c)(b+d))^2 \geq 4(a+c)(b+d)$
 $(a+b+c+d)^2 \geq 4(ab+bc+cd+da)$ 2

(b) (i) $f(x) = x^2 + ux + v = 0$ has one root between 0 and 1.
 $f(0) = v$
 $f(1) = 1 + u + v$
 As there is 1 root between 0 and 1, $f(x)$ changes sign between 0 and 1.
 $f(0)$ and $f(1)$ are of opposite sign

Sign
 $f(0) \cdot f(1) < 0$
 $v(1+u+v) < 0$ 2

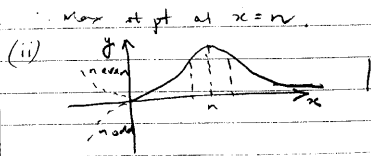
(ii) $\frac{1}{x+2} + \frac{u}{x+1} + \frac{v}{x} = 0$
 $x(x+1) + u(x+2) + v(x+2)(x+1) = 0$
 $x^2 + x + ux^2 + 2ux + vx^2 + 3vx + 2v = 0$
 $x^2(1+u+v) + x(1+2u+3v) + 2v = 0$

For 2 distinct roots $\Delta > 0$
 $\Delta = (1+2u+3v)^2 - 4(1+u+v)2v$
 $= 1 + 4u + 8v - 8v = 1 + 4u$
 $= 1 + 4u + 8v$
 > 0

$\therefore$ 2 distinct real roots
 Product of roots $= 2v(1+u+v)$
 < 0 3
 $\therefore$ Roots are of opposite sign
 $\therefore$ Only one root is positive

(c) $f(x) = x^n e^{-x}$
 (i) $f'(x) = e^{-x} \cdot nx^{n-1} + x^n \cdot -e^{-x}$
 $= x^{n-1} e^{-x} (n - x)$
 For st. pt $f'(x) = 0$
 $x = 0$ or $x = n$
 As $x > 0$, st. pt at $x = n$
 $f'(n-\epsilon) = +ve, +ve, +ve$
 > 0
 $f'(n+\epsilon) = +ve, +ve, -ve$
 < 0 3

x	$n-\epsilon$	n	$n+\epsilon$
$f'(x)$	+ve	0	-ve
Graph	/	-	\



(iii) $f(n-1) < f(n)$
 $(n-1)^n e^{-(n-1)} < n^n e^{-n}$
 $e < (\frac{n}{n-1})^n = (\frac{n}{n-1})^n = (1 + \frac{1}{n-1})^n$
 $f(n) > f(n+1)$
 $n^n e^{-n} > (n+1)^n e^{-(n+1)}$
 $e > (\frac{n+1}{n})^n = (1 + \frac{1}{n})^n$ 2
 $\therefore (1 + \frac{1}{n})^n < e < (1 + \frac{1}{n-1})^n$